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BIOLOGY

1. In which of the following exoskeleton of glass like silica is found:
A. Molluscs
B. Arthropods
C. Echinoderms
D. Starfish

2. The reserved CO2 and H2O are stored in:
A. Plastids
B. Mitochondria
C. Ribosomes
D. Nucleus

3. Which of the following replaced by thymine:
A. Uracil
B. Cytosine
C. Guanine
D. Adenosine

4. Which of the following is an enzyme
A. Polymerase
B. Hydroserae
C. Isomerise
D. Enzymate

5. Which of the following is the most efficient way of respiration in frog?
A. Pulmonary
B. Bucco-pharyngeal
C. Both bucco and pharyngeal
D. Cutaneous

6. Plants convert solar energy into chemical energy present in the bonds chemical substances namely:
A. NADPH2
B. Nucleotide
C. ATP
D. RNA

7. The hormone which helps in the contraction of uterus at time of birth is:
A. Prolactin
B. Oxytocin
C. Gonadotropin
D. Antidiuretic hormone

8. Subcutaneous fat accumulation is essential for:
A. Seal
B. Shark
C. Elephant
D. Kangaroo

9. Hirudin anti clotting factor is found in:
A. Leech
B. Earthworm
C. Waucheria
D. Neries

10. Animals and plants characters are found in:
A. Euglena
B. Paramecium
C. Bacteria
D. Virus

11. The photogenic bacteria transfer DNA material to another bacteria by:
A. Transduction
B. Transformation
C. Conjugation
D. Fragmentation

12. When a prosthetic group join with enzyme it forms:
A. Co-factor
B. Co-Enzyme
C. Holoenzyme
D. Apoenzyme

13. What doesn't take place in meiosis?
A. Chromosomes multiplied
B. Separation of chromosomes
C. Production of hormones
D. Production of enzymes

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14. When phenotypically tall plant is crossed with pure dwarf plant what cross is this:
A. Test
B. Monohybrid
C. Dihybrid
D. Multiple

15. In chick embryo the space between 2 layers is :
A. Coelome
B. Blastocoel
C. Primitive streak
D. Gastrocoel

16. In which region organic material and shells are found?
A. Littoral
B. Abassyal
C. Euphotic
D. Bathyal

17. Which parasite can survive in artificial medium:
A. Endoparasite
B. Ectoparasite
C. Obligate
D. Facultative

18. In the light antennae complex _____ is not found.
A. Hb
B. Chlorophyll a
C. Chlorophyll b
D. Carotenoid

19. Of the level of CO2 and H+ ions increases in the body the rate of breathing will:
A. Increase
B. Decrease
C. Remain constant
D. None

20. Lacteals require which muscle to flow.
A. Unstriated and involuntary
B. Striated and voluntary
C. Striated and involuntary
D. Unstriated and voluntary

21. The theory of inheritance of acquired characters was proposed by:
A. Devaries
B. Darwin
C. Lamark
D. Weisman

22. Which is associated with long term memory?
A. Hippocampus
B. Hypothalamus
C. Brain stem
D. Pons

23. Pituitary gland is controlled by which of the following?
A. Adrenal cortex
B. ACTH
C. Hypothalamus
D. Cytoxin

24. Bacteria takes rDNA in presence of:
A. NaOH
B. HCl
C. Calcium chloride
D. NaCl

25. Centromere at the end of chromosome:
A. Metacentric
B. Aerocentric
C. Submetacentric
D. None of the above

26. In plant cell, cell wall and cell plate is the product of:
A. Golgi bodies
B. Mitochondria

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C. Endoplasmic reticulum
D. Plastids

27. Tissue fluid doesn't contain:
A. WBC
B. Glucose
C. Amino acid
D. Fibrinogen

28. In the cycle of Plasmodium which of the following does not affected
A. Sporozoid
B. Merozoid
C. Cryptozoid
D. Tropozoid

29. Which enzyme is used to cut DNA into small pieces?
A. DNA polymerase
B. Vector
C. Restriction enzyme
D. DNA Ligase

30. Number of chromosomes in human cell is:
A. 48
B. 46
C. 44
D. 56

CHEMISTY

31. Grignard reagent on the reaction with methanol gives:
A. Primary alcohol
B. Secondary alcohol
C. Ethane
D. Tertiary alcohol

32. A compound X is oxidized in the presence of K2Cr2O7 and conc. H2SO4 to form a compound Y which on reaction with _____ gives a sweet /pleasant smell.
A. Aldehyde
B. Ester
C. Ether
D. Acetone

33. Yotal pressure is equal to sum of partial pressure of all the individual gases due to following reason:
A. Collisions with the walls of container
B. Effective collision of all individual gases
C. Collision with bottom of the container
D. None of the above

34. Metallic character of Al^{+2} is greater than boron because:
A. Due to less electron population
B. The number of neighboring atoms in lattice increase
C. Both of the above
D. None of the above

35. $NaNO_3$ + Heat → $NaNO_2$ +O_2 In this reaction:
A. Kp=Kc
B. Kp>Kc
C. Kp<Kc
D. Kp=Ksp

36. Following are metals of group IIA:
A. Mg, Be, Sr, Ra
B. Be, Ra, Rb, Ba
C. Mg, Be, Ca, Sr
D. None of the above

37. When the reaction is in equilibrium:
A. Production is equal to reaction
B. Rete of forward reaction is equal to that of backward reaction
C. Products are not equal to reactants
D. All of the above

38. Hydrogen has ground state energy of:

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A. 1.5 eV
B. -13.6 eV
C. -0.8 eV
D. 23.3 eV

39. Hydrogen is not absorbed by:
A. CO
B. Ni
C. Pt
D. Pd

40. Bond energy is needed to know:
A. Nature of compound
B. Nature of ionic compound
C. Nature of reaction
D. Nature of bonding

41. 1 ounce presume of a gas at STP contains:
A. 224000 m^3
B. 22.4 m^3
C. 224400m^3
D. 0.0224m^3

42. Swollen tounge, eyes and mouth are caused by:
A. B1
B. B2
C. B6
D. B12

43. 6,6-Nylon is polymer of:
A. Vinyl chloride
B. Aldehyde
C. Adipic acid + hexamethylene diamine
D. Both A and B

44. The triad contain chlorine 35.5 and iodine 126.5; give the atomic mass of Bromine:
A. 80
B. 81
C. 72
D. 90

45. Anode mud in blistening of copper is :
A. Impurity left
B. Pure copper
C. Electrolyte copper sulphate
D. Blisten copper

46. Reforming is a process in:
A. Which smaller compounds change into large compounds
B. Straight chain into branch chains
C. Less volatile compound into high volatile compound
D. Branched chain into cyclic compound

47. In destructive distillation of coal all of the following are obtained EXCEPT:
A. Coal liquor
B. Coal tar
C. Colle
D. Coal gas

48. Azimuthal quantum number shows:
A. Orientation in space
B. Size of the orbital
C. Shape of the orbital
D. None of the above

49. In ethane which of the following is correct:
A. Its p orbital is hybridized
B. Its s orbital is hybridized
C. Its p and s both orbitals are hybridized
D. Its p and s both orbitals are unhybridized

50. In SN2 reaction the rate of reaction is dependant upon:
A. Nucleophile
B. Substrate
C. Electrophile
D. Both substrate and nucleophile

51. Unsaturated compounds are formed by the following:

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A. Elimination
B. Substitution
C. Nucleophilic substitution
D. Electrophilic substitution

52. Glucose and fructose are linked by _____ linkage:
A. 1,2-glycosydic linkage
B. 2,2,4-glycosydic linkage
C. 1,4-glycosidic linkage
D. 1,6-glycosidic linkage

53. Compound having different forms; this statement is related to :
A. Allotropy
B. Isomerism
C. Polymorphism
D. Isomorphism

54. The heat absorbed at constant volume is equal to :
A. External pressure
B. Energy change
C. Internal pressure
D. All of the above

55. Equilibrium is established when:
A. Rate of forward reaction = rate of backward reaction
B. Rate of forward reaction > rate of backward reaction
C. Rate of forward reaction < Rate of backward reaction
D. Both B and C

56. Rate if reaction depends upon all of the following EXCEPT:
A. Concentration
B. Pressure
C. Temperature
D. Molecularity

57. Products of exothermic reaction are:
A. Unstable
B. Sometimes stable and sometimes unstable
C. Stable
D. None of the above

58. Cathode does not depend upon:
A. Neither on gas nor the material of the Cathode
B. Neither on H2SO4 nor on the liquid used
C. Both A and B
D. None of the above

59. The electronic configuration of noble gas is:
A. ns^2, np^5
B. ns^2, np^6
C. ns^2, np^6, nd^8
D. None of the above

60. Rutherford discovered that the nucleus of an atom has _____ charge.
A. No
B. Negative
C. Positive
D. Very less

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PHYSICS

61. If T if string is Increased 4 times what will the speed of wave?
A. V/2
B. V/4
C. 2V
D. 4V

62. What is the formula of convex lens without convention of sign:
A. $1/q - 1/q = 1/f$
B. $1/p + 1/q = 1/f$
C. $1/p - 1/p = 1/f$
D. $1/p - 1/q = 1/f$

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63. "Sone"bia the unit if loudness. What is the relation of sone with db?
A. 1 sone = 40db
B. 1 sone = 40db at 100Hz
C. 1 sone=m = 40db at 1000Hz
D. 1 sone = 40db at 10000Hz

64. Three sources each of 1.5 volt are connected in series what is net emf?
A. 3 volt
B. 4.5 volt
C. 1.5 volt
D. 6 volt

65. Electromagnetic wave depends upon:
A. Permittivity of medium only
B. Permeability of medium only
C. Both permittivity and permeability of medium
D. Nature of medium

66. If light enters from air to denser medium of "n" refractive index , wavelength will:
A. Increase "n" times
B. Decrease "n times
C. Decreases
D. Remains same

67. Shadow is formed due to phenomenon of:
A. Interference
B. Polarization
C. Diffraction
D. Both interference and diffraction

68. If A.B = 0 then angle between them is:
A. 0°
B. 45°
C. 90°
D. 60°

69. If power of a lens is 2 diopter, and of other is -4 diopter, what is focal length
Of combination of lenses:
A. 20cm
B. 50cm
C. 50cm
D. 100cm

70. Energy if hydrogen atom in ground state is:
A. -13.6 eV
B. -0.85 eV
C. -10.2 eV
D. -0.25 Ev

71. Dot Product of force and velocity:
A. Rate of Energy
B. Rate of work
C. Rate of power
D. Rate of change of momentum

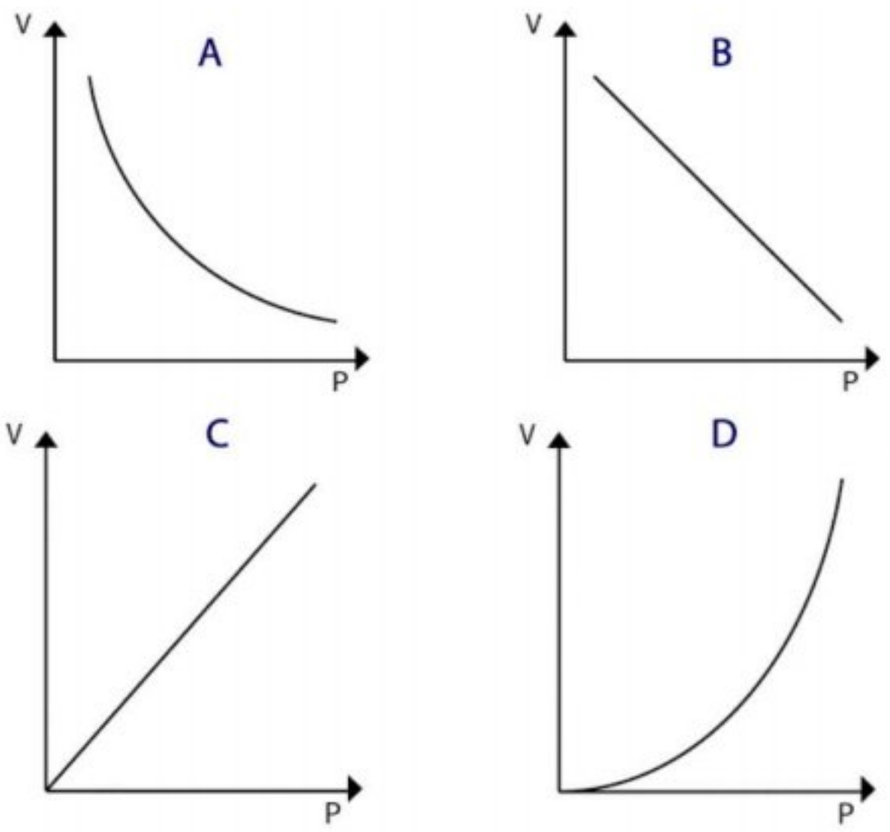
72. If "λ" is decay constant then half life is:
A. 0.693
B. 0.693/ λ
C. λ /0.693
D. 0.96/ λ

73. When a body moves downward I'm inclined plane in the presence of friction "f" , the force is:
A. $W \sin\theta - f$
B. $W \cos\theta - f$
C. $W \cos\theta$
D. $F - W \sin\theta$

74. Graph of Isobaric process is:

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75. According to thermodynamics, at constant volume Q=?

A. ΔU=ΔQ
B. ΔU=ΔW
C. ΔU=–ΔW
D. Q=0

76. Velocity of sound is:

A. Directly proportional to absolute temperature
B. Directly proportional to square root of absolute temperature
C. Inversely proportional to square root of absolute temperature
D. Directly proportional to pressure

77. If the axis of rotation passes through the body and the body moves around itself then motion is called:

A. Spin motion
B. Orbital motion
C. Rectilinear motion
D. Tranlatory motion

78. If radius of Earth decreases by half and mass remains unchanged, then the value of 'g' is:

A. 4g
B. 2g
C. g/4
D. g/2

79. Which instrument is used to detect the path of visible rays:

A. Geiger counter
B. Wilson cloud detector
C. Solid state detector
D. None of the above

80. Light having range of 7.6×10^{-6} cm to 4.0×10^{-7} cm lies in the region:

A. Infrared
B. X-rays
C. Visible
D. U.V Light

81. Relation between linear velocity and angular velocity:

A. $v = r\omega$
B. $r = v\omega$
C. $s = r$
D. $v = \Delta s / \Delta t$

82. Instrument having very high magnification used to see microganisms clearly:

A. Simple microscope
B. Compound microscope
C. Telescope
D. Magnifying glass

83. Gauss's can be used to find electric field intensity:

$v = r\omega$
$r = v\omega$
$s = r$
$v =$
Δs
Δt

A. Charge enclosed in a sphere
B. Charge placed between two opposed charged surfaces
C. Charge in an infinite sheet
D. All of the above

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84. Ratio of emf in a Potentiometer is directly proportional to Ratio of:
A. Balancing length
B. Balancing width
C. Balancing height
D. None of the above

85. Number of lines originating from q to the surface area of sphere is:
A. Flux intensity
B. Flux linkage
C. Flux density
D. Flux region

86 In solid state detector:
A. Sealed tube is used
B. Air alcohol and Argon is used
C. p-n junction is used
D. All of the above can be used

87. Velocity of alpha particle is:
A. $0.8x10^8 m/s$
B. $0.5x10^8 m/s$
C. $0.3x10^8 m/s$
D. $0.1x10^8 m/s$

88. 2nd harmonic is also known as:
A. Fundamental frequency
B. 1st overtone
C. 2nd overtone
D. Basic overtone

89. The carrier waves are over modulated to be distorted version of original modulating signal when:
A. ma=1
B. ma<1
C. ma>1
D. ma≠1

90. Frequency acquired to produce artificial gravity in satellite is:
A. 8.5 rev/min
B. 9.5 rev/min
C. 9.8 rev/min
D. 10.0 rev/min

ENGLISH

91. DESTITUTE (synonym):
A. Disabled
B. Isolated
C. Poor
D. Emangled

92. ORTHODOX (Antonym)
A. Emangled
B. Unconventional
C. Conventional
D. Traditional

93. ECONOMICAL (Antonym)
A. Maddest
B. Efficient
C. Extravagant
D. Expensive

94. LIMITATIONS (synonym)
A. Short comings
B. Restrictions
C. Vicinity
D. Inability

95. EXPEDIENT (synonym)
A. Scapegoat
B. Heterogeneous
C. Suitable
D. Apathetic

96. VORACIOUS (synonym)
A. Reject
B. Excitable
C. Honest
D. Greedy

97. HAMPER (antonym):
A. Ogle
B. Clarify

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C. Assist
D. Uncover

98. MONOTONOUS (synonym):
A. Limited
B. Conclusive
C. Extremist
D. Repeatedly

99. INTENSIVE (antonym)
A. Casual
B. Regressive
C. Progressive
D. Attentive

100. PRESICE (synonym):
A. Concise
B. Comprehensive
C. Exaggerate
D. Explanatory

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ANSWER KEY

Q NO	ANS KEY	Q NO	ANS KEY	Q NO	ANS KEY	NO	ANS KEY
1	C	26	A	51	A	76	B
2	A	27	D	52	A	77	A
3	A	28	D	53	C	78	A
4	D	29	C	54	B	79	B
5	A	30	B	55	A	80	C
6	C	31	C	56	D	81	A
7	B	32	B	57	C	82	B
8	B	33	A	58	A	83	D
9	A	34	D	59	B	84	A
10	A	35	B	60	C	85	C
11	A	36	C	61	C	86	C
12	C	37	B	62	B	87	D
13	C	38	B	63	C	88	B
14	C	39	A	64	B	89	C
15	A	40	D	65	C	90	B
16	B	41	B	66	A	91	C
17	A	42	B	67	D	92	B
18	A	43	C	68	C	93	D
19	A	44	B	69	B	94	B
20	A	45	A	70	A	95	B
21	C	46	B	71	C	96	D
22	A	47	C	72	B	97	C
23	C	48	C	73	A	98	D
24	C	49	C	74	A	99	A
25	D	50	D	75	A	100	C

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